Phase A Implementation Checklist 📋

Backend Implementation Steps

Step 1: Update Existing Model Files 🔄

1.1 Update `backend/app/models/__init__.py`

- [ ] **ACTION**: Replace entire file with new imports
- [ ] **FILE**: Use "Updated Models - Phase A Foundation" artifact
- [ ] **RESULT**: Imports all new model classes

1.2 Update `backend/app/models/school.py`

- [ ] **ACTION**: Replace entire file content
- [ ] **FILE**: Use "Updated School and User Models" artifact (School class)
- [ ] **RESULT**: Adds new relationships to existing School model

1.3 Update `backend/app/models/user.py`

- [ ] **ACTION**: Replace entire file content
- [ ] **FILE**: Use "Updated School and User Models" artifact (User class)
- [ ] **RESULT**: Adds parent_profile relationship and helper methods

1.4 Update `backend/app/models/student.py`

- [ ] **ACTION**: Replace entire file content
- [ ] **FILE**: Use "Student and Academic Record Models" artifact (Student class only)
- [ ] **RESULT**: Adds new fields and relationships to existing Student model

Step 2: Create New Model Files ✨

2.1 Academic Year Model

- [ ] **CREATE**: `backend/app/models/academic_year.py`
- [ ] **FILE**: Use "Academic Year Model" artifact
- [ ] **RESULT**: New AcademicYear model with validation methods

2.2 Subject Model

- [ ] **CREATE**: `backend/app/models/subject.py`
- [ ] **FILE**: Use "Subject Model" artifact
- [ ] **RESULT**: New Subject model with grade band and type classification

2.3 Room Model

- [ ] **CREATE**: `backend/app/models/room.py`
- [ ] **FILE**: Use "Room Model" artifact

- [ ] **RESULT**: New Room model with features and booking support

2.4 Classroom Models

- [ ] **CREATE**: `backend/app/models/classroom.py`
- [ ] **CREATE**: `backend/app/models/classroom_teacher_assignment.py`
- [ ] **CREATE**: `backend/app/models/teacher_role_template.py`
- [ ] **FILE**: Use "Classroom and Teacher Assignment Models" artifact
- [ ] **RESULT**: Three new models for classroom management

2.5 Student Academic Records

- [ ] **CREATE**: `backend/app/models/student_academic_record.py`
- [ ] **FILE**: Use "Student and Academic Record Models" artifact (StudentAcademicRecord class)
- [ ] **RESULT**: New model for tracking student progression

2.6 Special Needs Models

- [ ] **CREATE**: `backend/app/models/special_needs_tag_library.py`
- [ ] **CREATE**: `backend/app/models/student_special_need.py`
- [ ] **FILE**: Use "Special Needs Models" artifact
- [ ] **RESULT**: Two models for special needs management

2.7 Parent Models

- [ ] **CREATE**: `backend/app/models/parent.py`
- [ ] **CREATE**: `backend/app/models/parent_student_relationship.py`
- [ ] **FILE**: Use "Parent and Relationship Models" artifact
- [ ] **RESULT**: Two models for parent management

2.8 Enrollment Model

- [ ] **CREATE**: `backend/app/models/enrollment.py`
- [ ] **FILE**: Use "Enrollment Model" artifact
- [ ] **RESULT**: New model for student-classroom assignments

Step 3: Database Migration 🗄️

3.1 Create Migration File

- [ ] **CREATE**: `backend/alembic/versions/phase_a_foundation.py`
- [ ] **FILE**: Use "Phase A Migration File" artifact
- [ ] **RESULT**: Creates all new database tables and indexes

3.2 Run Migration

- [ ] **COMMAND**: `cd backend && python -m alembic upgrade head`
- [ ] **RESULT**: Database schema updated with all Phase A tables

Step 4: Create Schema Files 📝

4.1 New Schema Files

- [ ] **CREATE**: `backend/app/schemas/academic_year.py`
- [ ] **CREATE**: `backend/app/schemas/subject.py`
- [ ] **CREATE**: `backend/app/schemas/room.py`
- [ ] **CREATE**: `backend/app/schemas/classroom.py`
- [ ] **CREATE**: `backend/app/schemas/teacher_assignment.py`
- [ ] **CREATE**: `backend/app/schemas/special_needs.py`
- [ ] **CREATE**: `backend/app/schemas/parent.py`
- [ ] **FILE**: Use "Phase A Pydantic Schemas" artifact
- [ ] **RESULT**: Pydantic schemas for API validation

4.2 Update Student Schema

- [ ] **UPDATE**: `backend/app/schemas/student.py`
- [ ] **FILE**: Use "Phase A Pydantic Schemas" artifact (StudentOut class)
- [ ] **ACTION**: Replace existing student schema with updated version

Step 5: Create API Router Files 🛤️

5.1 Academic Years Router

- [ ] **CREATE**: `backend/app/routers/academic_years.py`
- [ ] **FILE**: Use "Phase A API Endpoints" artifact (academic_years.py section)
- [ ] **RESULT**: CRUD endpoints for academic year management

5.2 Subjects Router

- [ ] **CREATE**: `backend/app/routers/subjects.py`
- [ ] **FILE**: Use "Phase A API Endpoints" artifact (subjects.py section)
- [ ] **RESULT**: CRUD endpoints for subject management

5.3 Additional Routers

- [ ] **CREATE**: `backend/app/routers/rooms.py`
- [ ] **CREATE**: `backend/app/routers/special_needs.py`
- [ ] **CREATE**: `backend/app/routers/parents.py`
- [ ] **FILE**: Use "Missing Router Files" artifact
- [ ] **RESULT**: Complete API coverage for all new models

5.4 Update Classrooms Router

- [ ] **UPDATE**: `backend/app/routers/classrooms.py`
- [ ] **FILE**: Use "Phase A API Endpoints" artifact (classrooms.py section)
- [ ] **ACTION**: Replace existing classroom router with updated version

Step 6: Update Main Application 🚀

6.1 Update Main App File

- [ ] **UPDATE**: `backend/app/main.py`
- [ ] **FILE**: Use "Updated Main App and Route Includes" artifact (main.py section)
- [ ] **ACTION**: Add new router imports and includes

6.2 Update Router Init

- [ ] **UPDATE**: `backend/app/routers/__init__.py`
- [ ] **FILE**: Use "Updated Main App and Route Includes" artifact
- [ ] **ACTION**: Update router module documentation

Frontend Implementation Steps

Step 7: Update Frontend Components 🎨

7.1 Update Admin Interface

- [ ] **UPDATE**: `frontend/src/pages/Admin.jsx`
- [ ] **FILE**: Use "Updated Admin UI Components" artifact
- [ ] **ACTION**: Replace existing admin page with enhanced version
- [ ] **RESULT**: New tabs for academics, facilities, enhanced user management

7.2 Update Authentication Context

- [ ] **UPDATE**: `frontend/src/AuthContext.jsx`
- [ ] **FILE**: Use "Updated Main App and Route Includes" artifact (AuthContext section)
- [ ] **ACTION**: Add academic year context to auth state
- [ ] **RESULT**: Academic year available throughout app

Step 8: Seed Test Data 🌱

8.1 Create Seed Script

- [ ] **CREATE**: `backend/scripts/seed_phase_a_data.py`
- [ ] **FILE**: Use "Phase A Data Seeding Script" artifact
- [ ] **RESULT**: Comprehensive test data for development

8.2 Run Seed Script

- [ ] **COMMAND**: `cd backend && python scripts/seed_phase_a_data.py`
- [ ] **RESULT**: Database populated with realistic test data

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Verification Steps ✅

Step 9: Test the Implementation

9.1 Backend Health Check

- [ ] **START**: `cd backend && python -m uvicorn app.main:app --reload`
- [ ] **VERIFY**: Visit http://localhost:8000/health
- [ ] **VERIFY**: Visit http://localhost:8000/docs (API documentation)
- [ ] **RESULT**: Backend running with all new endpoints

9.2 Frontend Integration

- [ ] **START**: `cd frontend && npm run dev`
- [ ] **VERIFY**: Login with admin@springfield.edu / admin123
- [ ] **VERIFY**: Navigate through new admin tabs
- [ ] **RESULT**: Full system integration working

9.3 Test Key Features

- [ ] **TEST**: Create new academic year
- [ ] **TEST**: Add custom subject
- [ ] **TEST**: Create classroom with teachers
- [ ] **TEST**: View student enrollments
- [ ] **TEST**: Assign special needs to student
- [ ] **RESULT**: All Phase A features functional

File Structure After Implementation 📁

```
backend/
├── app/
│   ├── models/
│   │   ├── __init__.py                # 🖍️ UPDATED
│   │   ├── user.py                    # 🖍️ UPDATED
│   │   ├── school.py                  # 🖍️ UPDATED
│   │   ├── student.py                 # 🖍️ UPDATED
│   │   ├── academic_year.py           # ✨ NEW
│   │   ├── subject.py                 # ✨ NEW
│   │   ├── room.py                    # ✨ NEW
│   │   ├── classroom.py               # ✨ NEW
│   │   ├── classroom_teacher_assignment.py  # ✨ NEW
│   │   ├── teacher_role_template.py   # ✨ NEW
│   │   ├── student_academic_record.py # ✨ NEW
│   │   ├── special_needs_tag_library.py  # ✨ NEW
│   │   ├── student_special_need.py    # ✨ NEW
│   │   ├── parent.py                  # ✨ NEW
│   │   ├── parent_student_relationship.py  # ✨ NEW
│   │   └── enrollment.py              # ✨ NEW
│   ├── routers/
│   │   ├── main.py                    # 🖍️ UPDATED
│   │   ├── classrooms.py              # 🖍️ UPDATED
│   │   ├── academic_years.py          # ✨ NEW
│   │   ├── subjects.py                # ✨ NEW
│   │   ├── rooms.py                   # ✨ NEW
│   │   ├── special_needs.py           # ✨ NEW
│   │   └── parents.py                 # ✨ NEW
│   └── schemas/
│       ├── student.py                 # 🖍️ UPDATED
│       ├── academic_year.py           # ✨ NEW
│       ├── subject.py                 # ✨ NEW
│       ├── room.py                    # ✨ NEW
│       ├── classroom.py               # ✨ NEW
│       ├── teacher_assignment.py      # ✨ NEW
│       ├── special_needs.py           # ✨ NEW
│       └── parent.py                  # ✨ NEW
├── alembic/versions/
│   └── phase_a_foundation.py          # ✨ NEW
└── scripts/
    └── seed_phase_a_data.py           # ✨ NEW

frontend/src/
├── AuthContext.jsx                    # 🖍️ UPDATED
└── pages/
    └── Admin.jsx                      # 🖍️ UPDATED
```

Legend: ✨ NEW = Create new file | 🖍️ UPDATED = Modify existing file

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Success Criteria 🎯

After completing all steps, you should have:

- [ ] **24 new database tables** with proper relationships
- [ ] **60+ API endpoints** for complete CRUD operations
- [ ] **Enhanced admin interface** with 5 management sections
- [ ] **Test data** with realistic school, students, teachers, parents
- [ ] **Full user authentication** with role-based access
- [ ] **Academic year context** throughout the application
- [ ] **Special needs management** system
- [ ] **Multi-teacher classroom** assignments
- [ ] **Parent portal foundation** with granular permissions

Estimated Implementation Time: 2-3 hours (depending on familiarity with codebase)

Ready to build the future of school information systems! 🚀